

The Utility Problem

Voltage power quality monitoring is the process of measuring and analyzing the voltage level of an electrical system. The goal of voltage power quality monitoring is to ensure and report that the voltage level supplied to electrical devices and systems is within an acceptable range and that it does not experience any fluctuations or disturbances. Utilities might have their own power quality standards, although, there are some established standards like for example EN 50160 (for Europe).

Exceeding the voltage limit or not fulfilling certain parameters can cause damage to electrical equipment, and appliances, it can also cause operational problems and can shorten the lifespan of the devices connected to the grid. It can also cause fires and other hazards. Furthermore, bad power quality has an impact on end consumers of the electricity and can damage consumers' equipment too. To prevent and mitigate these issues, voltage levels should be closely monitored and any issues that arise should be addressed quickly.

The primary source of voltage power quality data is power quality monitoring equipment. There are several types of such equipment. By taking advantage of these devices, power quality engineers can detect and diagnose power quality issues. However, without the proper use of this measured data, 100% of the benefits of the data are not obtained.

Some of the benefits of data monitoring of voltage include:

1. Improved equipment and system reliability - by identifying and addressing power quality issues, data monitoring can help to reduce downtime and prolong the lifespan of electrical equipment and systems.
2. Compliance - by monitoring voltage levels, data monitoring can help to ensure compliance with local and national regulations and standards.
3. Safety - by identifying and solving power quality problems, data monitoring can help reduce high-voltage accidents.
4. Cost savings - by identifying and addressing power quality issues, data monitoring can help to reduce maintenance costs and prolong the lifespan of electrical equipment and systems.

Monitoring equipment exists for measurements of voltage power quality, but if data is siloed and lacks the context of other data elements, its usefulness can be limited. For these reasons, it is important to have a system which is integrating and contextualizing data from different sources, which then allows for a more comprehensive analysis of the electrical system. Such activity is then leading to better diagnosis and problem-solving, improved system reliability, better compliance, enhanced safety, and cost savings.

The Data Solutions and Methods

As an example, if we take into account standard EN 50160, this standard contains several aspects that need to be checked for power quality compliance. Awesense Platform can evaluate and visualize all these aspects. Power quality flags are sent to the Awesense Platform when a certain set parameter is exceeded. These parameters might be set directly in the Awesense True Grid Intelligence (TGI) software or the evaluation might be also done at the sensor level. If the flags are implemented in a near real-time fashion, the alerts can also be displayed in near real time. With the Awesense Energy Transition Platform, it is very easy to deploy Voltage Power Quality Monitoring in the entire grid.

Required data:

1. Power quality evaluation sensor data with time stamps (flags).

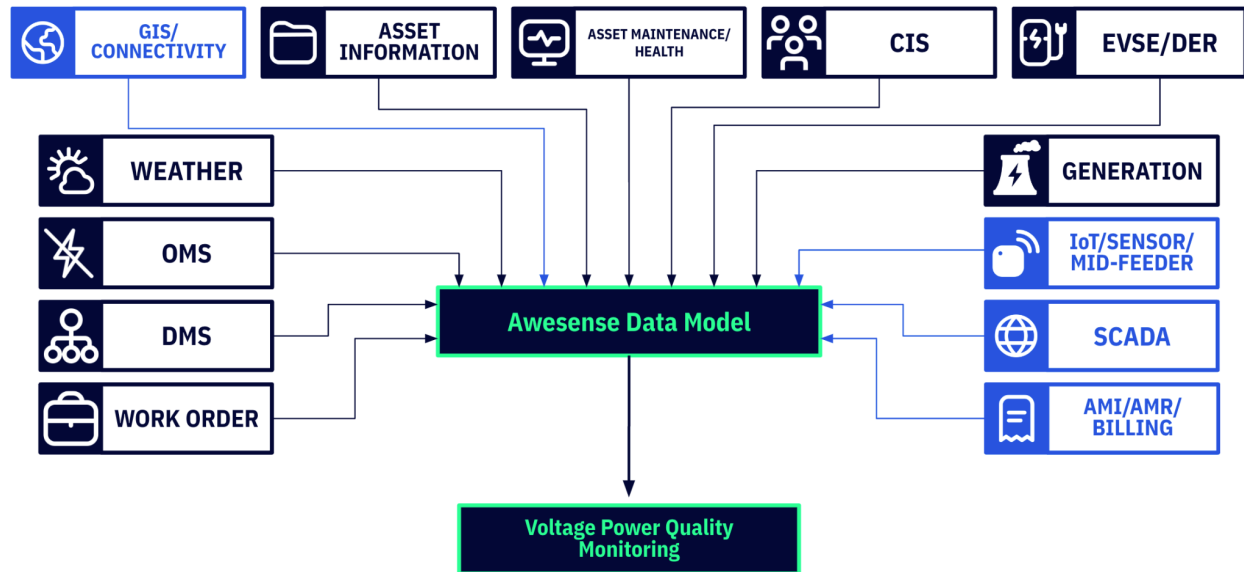
The Awesense Energy Transition Platform offers visualization using True Grid Intelligence (TGI) software, and alerts can be displayed directly in this interface. The alerts are displayed:

- In the tabular form where all attributes of the alert are listed. Additional information includes data as such timestamp, element ID or the phase where the threshold has been exceeded;
- On a map with a geospatial location, where the size of the circles on the map indicates the number of alerts at a location.

Awesense's TGI offers an easy, manageable and systematic approach to setting voltage alerts to detect the issues in the distribution grid immediately as they happen.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Grid Alerts - Please speak to Awesense about access to the sandbox environment.